

RATIONAL NUMBER

Warm-up

14. Điền các số hữu tỉ thích hợp vào ô trống :

Fill in the empty boxes with appropriate numbers

$-\frac{1}{32}$	$\times$	4	=	
:		$\times$		:
-8	:	$-\frac{1}{2}$	=	
=		=		=
	$\times$		=	

Vocabulary

English	IPA	Vietnamese	Example / How to read
rational number	/ˈræʃənəl ˈnʌmbə(r)/	số hữu tỉ	$-\frac{3}{4}$ Read: minus three over four
number line	/ˈnʌmbə(r) laɪn/	trục số	$-\frac{1}{2} < 0 < \frac{1}{2}$ Read: minus one over two is less than zero; zero is less than one over two
fraction	/ˈfrækʃən/	phân số	$\frac{7}{8}$ Read: seven over eight
numerator	/ˈnjuːməreɪtə(r)/	tử số	$\frac{5}{8}$ Read: the fraction, the numerator is five, the denominator is eight
denominator	/dɪˈnɒmɪneɪtə(r)/	mẫu số	$\frac{5}{8}$ Read: the fraction, the numerator is five, the denominator is eight
common denominator	/ˈkɒmən dɪˈnɒmɪneɪtə(r)/	mẫu số chung	$\frac{1}{2} = \frac{3}{6}, \frac{1}{3} = \frac{2}{6}$ Read: one over two equals three over six; one over three equals two over six
opposite number	/ˈɒpəzɪt ˈnʌmbə(r)/	số đối	$\frac{2}{5} + \left(-\frac{2}{5}\right) = 0$ Read: two over five plus open parenthesis minus two over five close parenthesis equals zero
reciprocal	/rɪˈsɪprəkəl/	số nghịch đảo	$\frac{2}{3} \times \frac{3}{2} = 1$ Read: two over three times three over two equals one

English	IPA	Vietnamese	Example / How to read
power	/ˈpaʊə(r)/	lũy thừa	$\left(-\frac{1}{2}\right)^4$ <i>Read: open parenthesis minus one over two close parenthesis raised to the fourth power</i>
exponent	/ɪkˈspəʊnənt/	số mũ	3^5 <i>Read: three raised to the fifth power</i>
equivalent fraction	/ɪˈkwɪvələnt ˈfrækʃən/	phân số bằng nhau	$\frac{1}{2} = \frac{2}{4}$ <i>Read: one over two equals two over four</i>
order of operations	/ˈɔːdə(r) əv ˌɒpəˈreɪʃənz/	thứ tự thực hiện phép tính	$2 + 3 \times 4 = 14$ <i>Read: two plus three times four equals fourteen</i>
increasing order	/ɪnˈkriːsɪŋ ˈɔːdə(r)/	thứ tự tăng dần	$-1 < 0 < \frac{1}{2}$ <i>Read: minus one is less than zero; zero is less than one over two</i>
decreasing order	/dɪˈkriːsɪŋ ˈɔːdə(r)/	thứ tự giảm dần	$\frac{3}{4} > 0 > -\frac{1}{2}$ <i>Read: three over four is greater than zero; zero is greater than minus one over two</i>
calculate	/ˈkælkjuleɪt/	tính toán	$\frac{5}{8} - \frac{1}{4} = \frac{3}{8}$ <i>Read: five over eight minus one over four equals three over eight</i>
equation	/ɪˈkweɪʒən/	phương trình	$x + \frac{1}{2} = \frac{3}{4}$ <i>Read: x plus one over two equals three over four</i>

Quick Review

A rational number is a number that can be written as

$$\frac{a}{b}, \quad a, b \in \mathbb{Z}, \quad b \neq 0.$$

The set of rational numbers is denoted by $\mathbb{Q}$.

The opposite of x is $-x$. Their sum is 0.

If $x \neq 0$, the reciprocal of x is $\frac{1}{x}$. In particular, the reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$ when $a \neq 0$.

Rational numbers can be compared by using a common denominator, changing them to decimals, or locating them on a number line. A number farther to the right is greater.

For addition and subtraction of fractions, use a common denominator. For division, multiply by the reciprocal of the divisor.

For $n \in \mathbb{N}$,

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

In a mixed calculation, use the order

brackets $\rightarrow$ powers $\rightarrow$ multiplication/division $\rightarrow$ addition/subtraction.

Exercises

Question 1. Copy and complete each set of equivalent fractions.

$$\text{a) } -3 = \frac{-3}{1} = \frac{\boxed{}}{4}$$

$$\text{b) } \frac{5}{8} = \frac{10}{\boxed{}} = \frac{\boxed{}}{40}$$

$$\text{c) } 0.6 = \frac{6}{10} = \frac{\boxed{}}{5}$$

$$\text{d) } -\frac{7}{9} = \frac{-14}{\boxed{}} = \frac{\boxed{}}{27}$$

Question 2. Write the correct symbol $<$, $>$, $=$ **in each box.**

$$\text{a) } -\frac{3}{4} \square -\frac{2}{3}$$

$$\text{b) } -0.4 \square -\frac{2}{5}$$

$$\text{c) } \frac{5}{8} \square 0.6$$

$$\text{d) } -1.25 \square -\frac{5}{4}$$

Question 3. Put the numbers in increasing order.

$$-\frac{5}{6}, \quad -0.72, \quad \frac{2}{3}, \quad 0, \quad -\frac{3}{4}, \quad 0.7.$$

Solution.

Question 4. Match each description with the correct number.

A. The opposite of $\frac{3}{5}$

$$1. \frac{4}{3}$$

B. The reciprocal of $\frac{3}{4}$

$$2. -\frac{3}{5}$$

C. A rational number equal to 0.75

$$3. \frac{3}{4}$$

D. A rational number equal to -1.2

$$4. -\frac{6}{5}$$

Question 5. Calculate. Show the main step in each calculation.

$$\text{a) } \frac{5}{8} - \frac{7}{12}$$

$$\text{b) } -\frac{4}{9} + \frac{5}{6}$$

$$\text{c) } \frac{7}{10} - \left(-\frac{3}{5}\right)$$

$$\text{d) } -\frac{11}{15} - \frac{2}{5}$$

Solution.

Question 6. Complete the working.

$$-\frac{7}{15} : \frac{14}{25} = -\frac{7}{15} \times \frac{\square}{\square} = -\frac{\square}{\square}.$$

Then calculate

$$\frac{5}{12} : \left(-\frac{25}{18}\right).$$

Solution.

Question 7. Match each expression with its value.

- | | |
|--|---------------------|
| A. $\left(-\frac{3}{4}\right)^2$ | 1. $-\frac{8}{125}$ |
| B. $\left(-\frac{2}{5}\right)^3$ | 2. $\frac{1}{2}$ |
| C. $\left(\frac{3}{7}\right)^5 : \left(\frac{3}{7}\right)^3$ | 3. $\frac{9}{16}$ |
| D. $\left(-\frac{1}{2}\right)^4 \cdot 2^3$ | 4. $\frac{9}{49}$ |

Question 8. Copy and complete the exponent laws.

- a) $\left(\frac{5}{6}\right)^3 \left(\frac{5}{6}\right)^4 = \left(\frac{5}{6}\right)^\square$
- b) $\left(-\frac{7}{9}\right)^8 : \left(-\frac{7}{9}\right)^5 = \left(-\frac{7}{9}\right)^\square$
- c) $\left(-\frac{2}{3}\right)^4 \left(-\frac{2}{3}\right)^3 : \left(-\frac{2}{3}\right)^5 = \left(-\frac{2}{3}\right)^\square = \frac{\square}{\square}.$

Question 9. Calculate using the correct order of operations.

- a) $\frac{2}{3} - \frac{3}{4} \cdot \frac{8}{9}$
- b) $\left(-\frac{5}{6} + \frac{1}{3}\right) : \frac{5}{4}$
- c) $1 - \left[\frac{2}{5} - \left(-\frac{3}{10}\right)\right]$
- d) $\left(-\frac{1}{2}\right)^3 + \frac{3}{4} : \left(-\frac{3}{2}\right)$

Solution.

Question 10. Find the mistake. A learner writes

$$-\frac{2}{3} + \frac{5}{6} = -\frac{4}{6} - \frac{5}{6} = -\frac{9}{6}.$$

Explain the mistake and write the correct calculation.

Solution.

Question 11. Choose the correct answer, then explain your choice.

$$A = \frac{3}{7} \cdot \frac{11}{5} + \frac{3}{7} \cdot \frac{4}{5}.$$

- | | |
|--------------------|------------------|
| A. $\frac{33}{35}$ | B. $\frac{9}{7}$ |
| C. $\frac{3}{5}$ | D. 3 |

Solution.

Question 12. Find numbers between two rational numbers.

Write **three different rational numbers** x such that

$$-\frac{2}{3} < x < -\frac{1}{2}.$$

Explain briefly how you know that each answer is correct.

Solution.

Question 13. Find x . Show all your working.

a) $x - \frac{5}{6} = -\frac{1}{4}$

c) $\frac{x + 1}{2} = \frac{5}{6}$

b) $\frac{3}{4}x = -\frac{9}{10}$

d) $\frac{x - 2}{3} + \frac{x + 1}{6} = 1$

Solution.

Question 14. For each statement, write **always true**, **sometimes true**, or **never true**. Give a reason or an example.

- a) The sum of two rational numbers is positive.
- b) The product of two negative rational numbers is positive.
- c) A rational number has a reciprocal.

Solution.

Question 15. Solve the real-life problems.

- a) At 6 a.m., the temperature is -3.5°C . By noon, it rises by 8.2°C . In the evening, it falls by 2.7°C . What is the temperature in the evening?
- b) A water tank is $\frac{3}{5}$ full. Then $\frac{1}{4}$ of the tank’s total capacity is used. What fraction of the tank remains full?

Solution.

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Question 16. Solve the real-life problem. A 3.5 m ribbon is cut into two pieces. The first piece is $\frac{3}{4}$ m long and the second piece is $1\frac{1}{4}$ m long. How much ribbon is left?

Solution.

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Bonus 1. Fill in the blank. Find the value of

$$\left(-\frac{3}{5}\right) \cdot \frac{25}{9} \cdot \left(-\frac{6}{5}\right).$$

Answer: _____.

Solution.

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Bonus 2. Fill in the blank. Let n be an integer. If

$$-\frac{3}{4} < \frac{n}{12} < -\frac{1}{2},$$

then all possible values of n are _____.

Solution.

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HOMEWORK

1. Complete the sentences using the words in the box.

numerator

denominator

equivalent fractions

reciprocal

exponent

increasing order

decreasing order

equation

a) In $\frac{5}{8}$, 5 is the _____ and 8 is the _____.

b) $\frac{1}{2}$ and $\frac{2}{4}$ are _____.

c) The _____ of $\frac{3}{5}$ is $\frac{5}{3}$.

d) In 2^5 , the number 5 is the _____.

e) $-\frac{3}{4} < 0 < \frac{1}{2}$ is written in _____.

f) $\frac{4}{5} > 0 > -\frac{2}{3}$ is written in _____.

g) $x + \frac{1}{2} = \frac{3}{4}$ is an _____.

2. Write how you would read each expression aloud in English. Use the forms from the Vocabulary table.

a) $-\frac{3}{4}$

b) $\frac{2}{3} \times \frac{3}{5} = \frac{2}{5}$

c) $\left(-\frac{1}{2}\right)^4$

d) $-\frac{2}{3} < -\frac{1}{2} < 0$

Solution.

3. Copy and complete.

a) $\frac{3}{4} = \frac{\boxed{}}{20}$

b) $-\frac{5}{6} = \frac{-15}{\boxed{}}$

c) $0.4 = \frac{\boxed{}}{5}$

d) $-2 = \frac{\boxed{}}{7}$

4. Write $<$, $>$, $=$ in each box, then write the four numbers in decreasing order.

$$-\frac{5}{8} \square -0.6, \quad \frac{7}{10} \square 0.7, \quad -\frac{2}{3} \square -\frac{3}{4}.$$

Numbers: $-\frac{5}{8}$, -0.6 , $\frac{7}{10}$, $-\frac{2}{3}$.

Solution.

5. Calculate. Show the main steps.

a) $-\frac{7}{12} + \frac{5}{8}$

b) $\frac{3}{5} - \left(-\frac{7}{10}\right)$

c) $-\frac{4}{9} \cdot \frac{15}{8}$

d) $\frac{5}{6} : \left(-\frac{25}{18}\right)$

Solution.

6. Calculate using powers and the correct order of operations.

a) $\left(-\frac{2}{3}\right)^4 : \left(-\frac{2}{3}\right)^2$

b) $1 - \frac{3}{4} \cdot \left(-\frac{2}{3}\right)$

Solution.

7. Find x . Show all your working.

a) $x + \frac{3}{5} = -\frac{1}{10}$

b) $\frac{2}{3}x = -\frac{4}{9}$

Solution.

8. A learner writes

$$\frac{1}{2} - \frac{3}{4} = \frac{1-3}{2-4} = 1.$$

Explain the mistake and give the correct answer.

Solution.

9. **Real-life problem.** A bottle contains 2.4 L of juice. During lunch, $\frac{3}{4}$ L is used. Later, another 0.65 L is used. How much juice remains in the bottle?

Solution.